

CLUSTERING

CLUSTER VALIDITY – PART III

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CLUSTER VALIDITY

The following concept will be introduced:

- ✓ **VALIDITY PARADIGM**



IDENTIFY THE CLUSTERING STRUCTURE
AND VALIDATION TYPE



EXTERNAL OR INTERNAL?

Both **EXTERNAL AND INTERNAL CRITERIA** are closely related to **statistical methods** and **hypothesis tests**.

The **VALIDITY PARADIGM**, for a clustering structure, is based on the following **NULL HYPOTHESIS**

"THERE IS NO STRUCTURE ON THE DATASET"



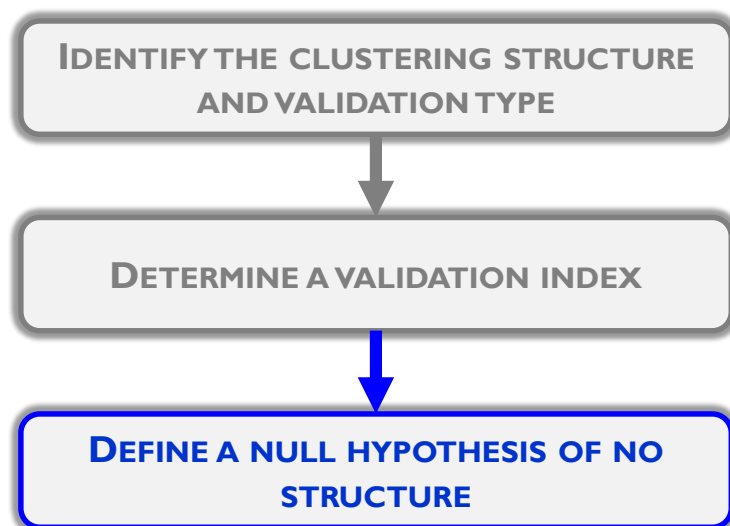
IDENTIFY THE CLUSTERING STRUCTURE
AND VALIDATION TYPE



DETERMINE A VALIDATION INDEX

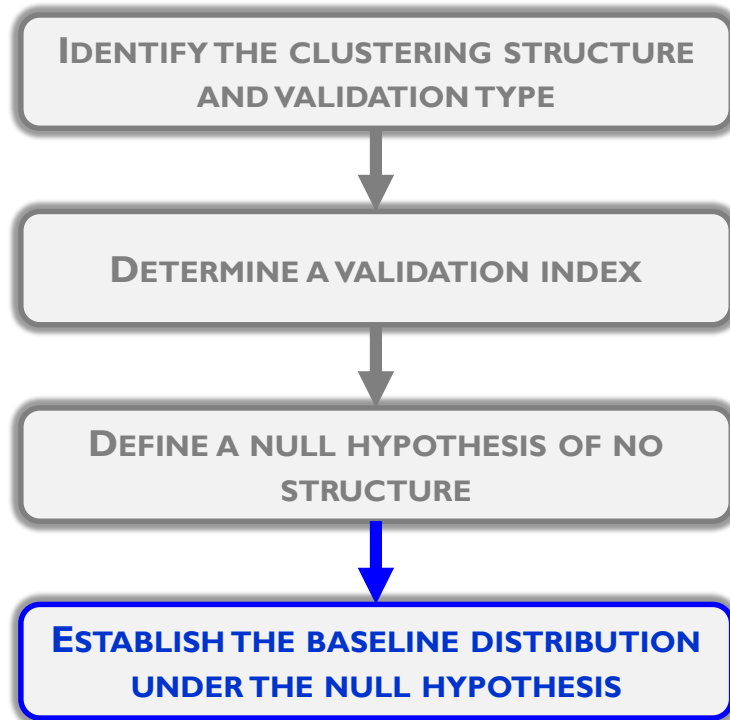


WHICH SPECIFIC VALIDATION INDEX?

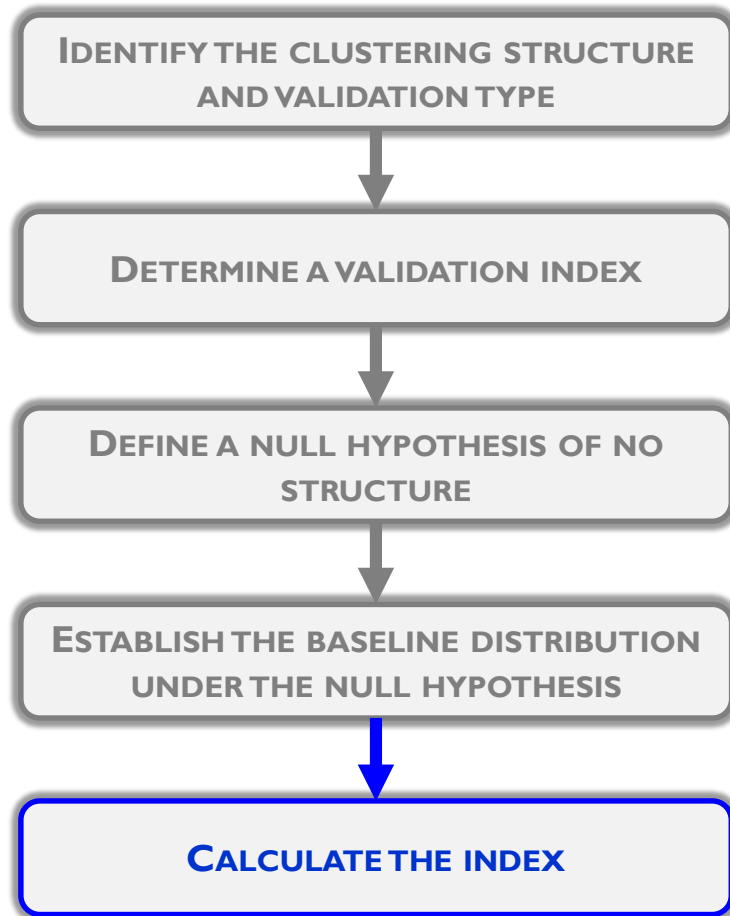


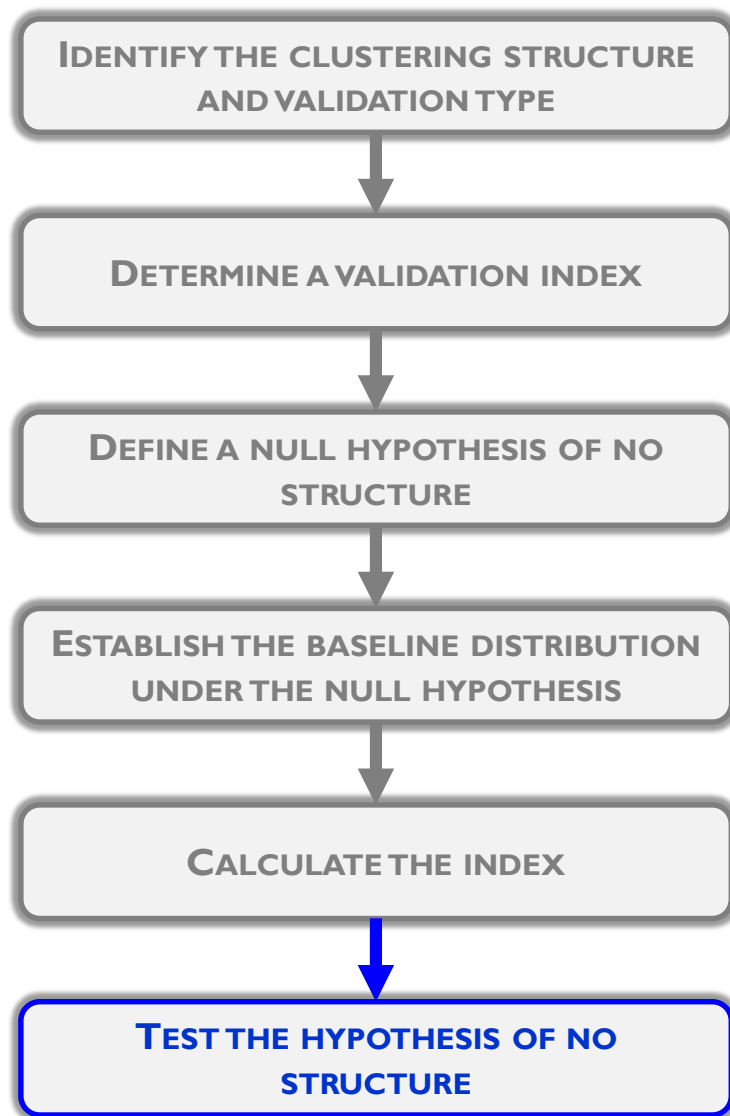
Three commonly used null hypotheses

- **RANDOM POSITION HYPOTHESIS**, all the locations of the m data points in some specific region of a n -dimensional space are equally likely
 - **RANDOM GRAPH HYPOTHESIS**, all $[m \times m]$ rank order proximity matrices are equally likely
 - **RANDOM LABEL HYPOTHESIS**, all permutations of the labels on m data points are equally likely
- FOR RATIO DATA** ←
- FOR ORDINAL PROXIMITIES BETWEEN PAIRS OF DATA OBJECTS** ←
- FOR ALL DATA TYPES** ←



**MONTÉ CARLO ANALYSIS
AND BOOTSTRAPPING**





You hope to reject the null hypothesis that there is no structure in the clustering you have developed